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#Date: 2026-09-13
#By: Divyam Sood
#Compare two numbers

.data
n1: .asciiz "Enter first number:"
n2: .asciiz "Enter second number:"
equall: .asciiz "both numbers are equal"
state: .asciiz " is greater than "

.globl main
.text

main:
    la $a0, n1
    li $v0, 4
    syscall
    li $v0, 6
    syscall
    mov.s $f1, $f0
    la $a0, n2
    li $v0, 4
    syscall
    li $v0, 6
    syscall
    #$f0 has second number

    #Check equality
    c.eq.s $f0, $f1
    bc1t equal
    c.lt.s $f0, $f1
    bc1t case_a #first number greater
    bc1f case_b #second number greater

case_b:
    mov.s $f12, $f0
    li $v0, 2
    syscall
    la $a0, state
    li $v0, 4
    syscall
    mov.s $f12, $f1
    li $v0, 2
    syscall
    beq $zero, $zero, end

case_a:
    mov.s $f12, $f1
    li $v0, 2
    syscall
    la $a0, state
    li $v0, 4
    syscall
    mov.s $f12, $f0
    li $v0, 2
    syscall
    beq $zero, $zero, end


equal:
    la $a0, equall
    li $v0, 4
    syscall
    beq $zero, $zero, end

end:
    # exit program
    li  $v0, 10         # system call code for exit = 10
    syscall             # call operating system
```